# Do Now · retrieval

## Lesson 125 · Assembly Language and the Little Man Computer

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is the purpose of a function return value?

### Q2

After pushing A then B onto a stack, which is popped first?

### Q3

What is the output of a full adder with A=1, B=1 and carry-in=0?

### Q4

Why is the program counter important in a stored-program computer?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Mailboxes — Instructions and numeric data occupy addressed memory mailboxes. DAT reserves a data location. Programs execute the instructions addressed by the program counter.

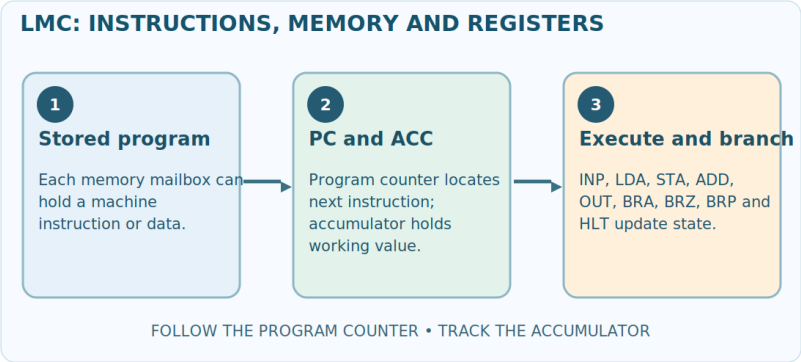
## Core idea 2

Accumulator and PC — INP reads into ACC; LDA reads memory into ACC; STA writes ACC to memory. The PC normally advances after a fetch, unless a branch selects a new address.

## Core idea 3

Branches and stopping — ADD and SUB update ACC; OUT prints ACC; BRA always branches; BRZ tests for zero; BRP tests a non-negative result in standard LMC; HLT ends the program.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.

# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

LMC — Educational model of a stored-program accumulator computer.

accumulator — Register holding current working data and arithmetic results.

program counter — Register holding address of the next instruction to fetch.

mnemonic — Symbolic name representing a machine instruction.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

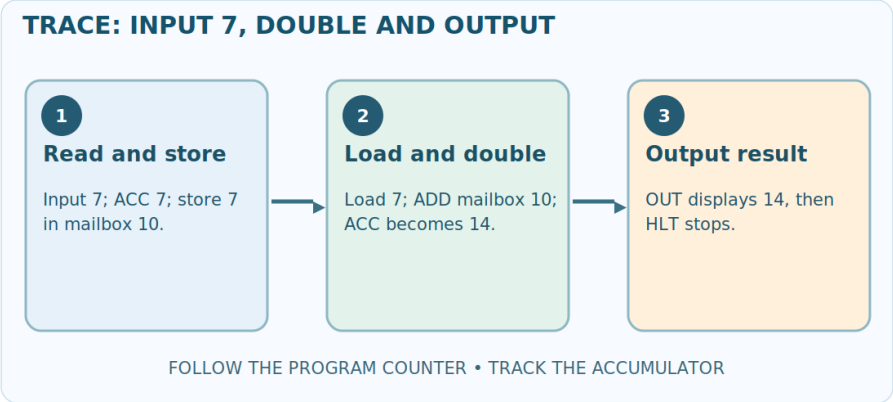
## Problem

Input is 7. Trace this program, including the accumulator, data mailbox and output.

## Follow the worked method

1. Track each change — 00 INP: ACC = 7. — 01 STA 10: mailbox 10 becomes 7; ACC remains 7. — 02 LDA 10: ACC = 7. — 03 ADD 10: ACC = 14. — 04 OUT: output is 14 . — 05 HLT: stop. Data mailbox 10 is separate from instruction addresses 00–05.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict the registers
What are ACC and mailbox 20 after the first two instructions?

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Explain what the PC, accumulator and memory mailboxes represent in LMC. [6]

## Question 2

Trace INP; STA 30; LDA 30; ADD 31; OUT; HLT, with input 5 and mailbox 31 containing 3. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Write LMC instructions to read an integer and output it, using a BRZ branch when zero. Explain how you would test both paths. [8]

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

In LMC, STA n loads the accumulator with the contents of mailbox n.

## Use key vocabulary from memory

1. Explain LMC in your own words.

2. Explain accumulator in your own words.

3. Explain program counter in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.